

PSFNet: Estimating the Modulated Transfer Function for Image Reconstruction

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Abstract

Computed tomography (CT) image quality depends critically on the reconstruction kernel chosen during image reconstruction, which balances spatial resolution against noise. Converting an image between kernels (e.g., smooth to sharp) currently requires re-reconstruction from raw projection data — data that is large, costly to store, and frequently unavailable. Existing image translation methods bypass this requirement but ignore an important property: the relationship between reconstructions under different kernels is governed by the modulation transfer function (MTF) in the frequency domain. We propose PSFNet, a network that estimates the MTF of reconstruction kernels directly from the frequency domain. Given a pair of images reconstructed with different kernels, PSFNet predicts both MTFs in the frequency domain and derives conversion filters from them, enabling image generation across kernels in a cyclic fashion. Evaluated against CycleGAN, Pix2Pix, CUT, and diffusion baselines, PSFNet achieves the highest SSIM (0.987), PSNR (43.2 dB) and the lowest FID (4.05) on the reconstruction benchmarks. On phantom image reconstructions, PSFNet achieves the highest SSIM(0.9) against ground truth, while the strongest baseline drops to approximately 0.4. PSFNet enables kernel conversion to be performed from reconstructed images alone, without access to raw projection data.

1 Introduction

A central property of a CT scan image is the modulation transfer function (MTF). In the Fourier domain, the reconstructed image is the product of the true signal spectrum and the kernel’s MTF. Knowledge of the MTF therefore makes it possible to compare acquisitions across scanners and protocols, to correct images for resolution loss, and to synthesize images as if they had been reconstructed with a different kernel — for example, converting a smooth-kernel image into its sharp-kernel counterpart, or vice versa.

Reconstructing an image with a different kernel is extremely impractical, since it requires re-running the reconstruction on the raw projection data. Raw data is substantially larger than the reconstructed images, expensive to store and transmit, and is frequently discarded after reconstruction. Access to it cannot be assumed in clinical or research settings. Image-domain filtering has been proposed as an alternative [1], in which the ratio of the two kernels’ modulation transfer functions is applied to the reconstructed image in the frequency domain. However, this requires the MTFs themselves to be known from explicit measurements, which are not routinely available. Kernel estimation from image frequency content has also been demonstrated in related imaging tasks: spectrum-to-kernel translation recovers blur kernels from the Fourier amplitude spectrum for blind image super-resolution [2], but targets natural-image degradation rather than CT reconstruction kernels. Estimating the MTF directly from a reconstructed image would eliminate this dependency entirely, enabling kernel conversion to be performed directly on the CT scan images.

In this paper, we propose PSFNet, a network that explicitly estimates the MTF of reconstruction kernels from reconstructed images. Given two images reconstructed with different kernels, PSFNet predicts their MTFs in the frequency domain, from which conversion filters are derived and used to generate the counterpart image in a cyclic fashion. Our contributions are:

- We formulate kernel estimation directly in the frequency domain, recovering the MTF rather than performing unconstrained image translation.
- We demonstrate consistent improvements over CycleGAN, Pix2Pix, CUT, and diffusion baselines on SSIM, FID, and PSNR, and validate generalization on phantom reconstructions.

2 Method

2.1 Theory

Considering two pairs of images reconstructed from two kernels (one from a smooth kernel k_1 and one from a sharp kernel k_2), $\{I_1^{k_1}, I_1^{k_2}\}$ and $\{I_2^{k_1}, I_2^{k_2}\}$. For each pair, the smooth and the sharp image can be represented as the convolution of the true underlying signals x_1 and x_2 with their respective kernels:

$$I_1^{k_1} = x_1 * k_1, \quad I_1^{k_2} = x_1 * k_2 \quad (1)$$

$$I_2^{k_1} = x_2 * k_1, \quad I_2^{k_2} = x_2 * k_2 \quad (2)$$

For each pair, the smooth image can be estimated from the sharp image in the frequency domain as:

$$I_1^{k_1} = F^{-1} \left(\frac{F(I_1^{k_2}) \cdot \text{MTF}_{k_1}}{\text{MTF}_{k_2} + \varepsilon} \right) \quad (3)$$

where F is the Fourier transform operator. MTF_{k_1} and MTF_{k_2} are the estimated modulated transfer functions for the reconstruction kernels k_1 and k_2 , respectively. The term ε is used to prevent division by zero in the numerator.

PSFNet estimates the MTF of a given image. From two isolated images, $\{I_1^{k_1}, I_2^{k_2}\}$, we have:

$$F(I_1^{k_1}) = F(x_1) \cdot \text{MTF}_{k_1} \quad (4)$$

$$F(I_2^{k_2}) = F(x_2) \cdot \text{MTF}_{k_2} \quad (5)$$

During training, we input the two isolated images $\{I_1^{k_1}, I_2^{k_2}\}$ to PSFNet one by one for each iteration. PSFNet generates MTF_{k_1} and MTF_{k_2} . The two generated MTFs are then input into the frequency-to-spatial module. The generated I_1 with PSF k_2 , denoted as $I_{1,\text{gen}}^{k_2}$, can be represented as:

$$I_{1,\text{gen}}^{k_2} = F^{-1} \left(\frac{F(I_1^{k_1}) \cdot \text{MTF}_{k_2}}{\text{MTF}_{k_1} + \varepsilon} \right) \quad (6)$$

Similarly, we can generate $I_{2,\text{gen}}^{k_1}$. The image reconstruction loss is formulated using the L1 norm to measure the pixel-wise difference between the generated images and their corresponding ground truths:

$$\mathcal{L}_{\text{ReconIMG}} = \|I_{1,\text{gen}}^{k_2} - I_1^{k_2}\|_1 + \|I_{2,\text{gen}}^{k_1} - I_2^{k_1}\|_1 \quad (7)$$

To further guide the model, we employ a model distillation approach in which a pre-trained teacher model, *FilterEstimator*, provides reference conversion filters ($F_{s \rightarrow sh}^{\text{teacher}}$ and $F_{sh \rightarrow s}^{\text{teacher}}$). Given the power spectral density (PSD) of a pair of images reconstructed with different kernels, *FilterEstimator* directly predicts the two 2D conversion filters, smooth-to-sharp and sharp-to-smooth.

$$F_{s \rightarrow sh} = \frac{k_{\text{sharp}}}{k_{\text{smooth}} + \epsilon}, \quad F_{sh \rightarrow s} = \frac{k_{\text{smooth}}}{k_{\text{sharp}} + \epsilon} \quad (8)$$

The filter estimation loss, $\mathcal{L}_{\text{FT}}$, is computed as the L1 distance between the predicted filters and the teacher filters:

$$\mathcal{L}_{\text{FT}} = \|F_{s \rightarrow sh} - F_{s \rightarrow sh}^{\text{teacher}}\|_1 + \|F_{sh \rightarrow s} - F_{sh \rightarrow s}^{\text{teacher}}\|_1 \quad (9)$$

The final objective function is a weighted sum of the filter loss and the image reconstruction loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{FT}} + \mathcal{L}_{\text{ReconIMG}} \quad (10)$$

2.2 Experimental Method

We performed experiments to validate our method against existing image translation techniques (CycleGAN, Pix2Pix, CUT, and Diffusion models). Evaluations were conducted on reconstructions of patient CT scan volumes using standard metrics (SSIM, FID, PSNR) as well as phantom reconstructions to evaluate generalization capabilities.

2.3 Pseudo Code for Training

Algorithm 1 Training procedure for PSFNet

- 1: **Input:** Training dataloader yielding $\{I_{\text{smooth},1}, I_{\text{sharp},1}, I_{\text{smooth},2}, I_{\text{sharp},2}\}$
 - 2: **Input:** Pre-trained Teacher Model $\mathcal{T}$, Student Model $\mathcal{S}$, λ_{recon}
 - 3: **for** each batch in dataloader **do**
 - 4: **PSD Computation**
 - 5: **Teacher Forward Pass (Frozen Weights)**
 - 6: **Student Forward Pass**
 - 7: **Image Generation**
 - 8: **Compute Losses (L1)**
 - 9: **Optimize Weights**
 - 10: **end for**
-

3 Results

The quantitative evaluation of our proposed MTF estimation methods against established image translation baselines is presented in Table 1. Our proposed frequency domain approach, which estimates the MTF from the frequency domain, consistently outperforms the baseline models across all evaluated metrics.

Specifically, the 256-point MTF configuration achieved the highest Structural Similarity Index (SSIM) for both smooth (0.9872) and sharp (0.9814) reconstructions. It also achieved the highest Peak Signal to Noise Ratio (PSNR) with 43.17 dB for smooth kernels and 42.37 dB for sharp kernels. Meanwhile, the Spline MTF method

yielded the best Fréchet Inception Distance (FID) scores, achieving 4.0539 for smooth and 3.8779 for sharp reconstructions, indicating superior perceptual quality. The 2D kernel approach also remained highly competitive, surpassing all non MTF baselines.

In contrast, baseline generative models struggled to maintain high-fidelity structural and perceptual accuracy. Pix2Pix performed the best among the baselines but still fell short of the MTF-based methods, particularly in smooth FID (11.1390) and sharp PSNR (40.82 dB). CycleGAN and CUT exhibited substantially lower SSIM scores, ranging between 0.86 and 0.91. The Diffusion model recorded the lowest quantitative performance overall, with an SSIM of 0.7806 (smooth) and 0.7545 (sharp), and the worst FID scores (12.3587 smooth, 15.4395 sharp).

Table 1: Quantitative evaluation across reconstruction models for smooth and sharp kernel targets.

Model	SSIM Smooth $\uparrow$	SSIM Sharp $\uparrow$	FID Smooth $\downarrow$	FID Sharp $\downarrow$	PSNR Smooth $\uparrow$	PSNR Sharp $\uparrow$
2D Kernel	0.9735	0.9690	6.9418	6.8485	41.54	41.53
256 Point MTF	0.9872	0.9814	4.2464	4.1389	43.17	42.37
Spline MTF	0.9757	0.9713	4.0539	3.8779	42.81	41.76
Pix2Pix	0.9767	0.9733	11.1390	5.0768	41.85	40.82
CycleGAN	0.8626	0.9015	11.1486	6.9894	35.90	37.29
CUT	0.9178	0.9046	11.0570	6.8679	37.98	37.16
Diffusion	0.7806	0.7545	12.3587	15.4395	30.21	29.88

Furthermore, qualitative and quantitative evaluations on phantom reconstructions corroborated our findings on the patient volumes. Visual inspection of the phantom reconstructions (Fig. 1) demonstrates that the MTF model produces images that closely match the ground truth. Baseline methods on the other hand, introduced noticeable artifacts and structural deviations.

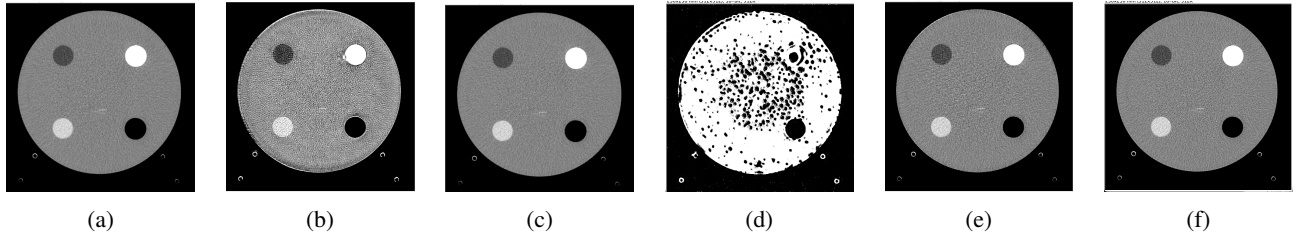


Figure 1: Phantom reconstructions: (a) ground truth, (b) CycleGAN, (c) PSFNet (MTF model), (d) CUT, (e) Pix2Pix, (f) reconstruction using the measured MTF.

4 Conclusion

We proposed PSFNet, a frequency-domain network that estimates the MTF of CT reconstruction kernels directly from reconstructed images, eliminating the need for raw projection data in kernel conversion. By combining knowledge distillation from a pre-trained teacher with cyclic image-reconstruction consistency, PSFNet recovers conversion filters that are both physically grounded and pixel-accurate. Evaluated against CycleGAN, Pix2Pix, CUT, and diffusion baselines, PSFNet achieved the highest SSIM (0.987), PSNR (43.2 dB), and FID (4.05).

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